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DIVISION OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTER SCIENCE AND TECHNOLOGY

A PROJECT BASED EVALUATION REPORT

SUBMITTED BY

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URK24CS1002

COURSE CODE
23CS2050

COURSE NAME

EMBEDDED SYSTEMS

APRIL 2026

INDUSTRIAL CERTIFICATION

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This professional has demonstrated initiative and a commitment to deepening their skills and advancing their career. Well done!

13th January 2026
Certificate code :9713327

Krishna Kumar
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Embedded Smart Pill Box with Missed-Dose Alert Mechanism

PROJECT REPORT

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APRIL 2026

ABSTRACT

The **Embedded Smart Pill Box with Missed-Dose Alert Mechanism** helps users take medication on time using an RTC-based reminder system. It provides alerts through a buzzer, LED, and LCD display at scheduled times. A touch sensor activates a servo motor to open the pill compartment, while an IR sensor detects whether the medicine has been taken. This low-cost and user-friendly system ensures both reminders and confirmation of medication intake, making it suitable for elderly users.

OBJECTS AND SCOPES:

- To design a cost-effective and user-friendly smart pill box system.
- To provide timely medication reminders using buzzer, LED, and LCD display.
- To automate pill box opening using a servo motor and touch sensor interaction.
- To detect whether medicine is taken using an IR sensor.
- To reduce missed doses and improve medication adherence.
- To develop a simple system without the need for mobile apps or internet.

Key Features and Functionalities:

- Provides time-based medication reminders using an RTC module
- Generates alerts through buzzer and LED indicators
- Displays time and reminders on an LCD screen
- Uses touch sensor for user interaction
- Servo motor automatically opens the pill compartment
- IR sensor detects whether the medicine is taken.
- Identifies missed doses for better monitoring

CHAPTER 1

INTRODUCTION

The **Embedded Smart Pill Box with Missed-Dose Alert Mechanism** is developed to help users take their medication on time and avoid missed doses. It uses an Arduino Uno with an RTC module to provide accurate time-based reminders through a buzzer, LED, and LCD display. A touch sensor allows the user to open the pill compartment using a servo motor, while an IR sensor detects whether the medicine has been taken. By combining alerts with detection, the system ensures both reminder and confirmation, making it a reliable and user-friendly solution for improving medication adherence.

Background Information about the Project:

Medication adherence is a critical aspect of healthcare, especially for elderly individuals and patients with chronic diseases. Many patients fail to take their medicines on time due to forgetfulness, busy schedules, or complex medication routines. Traditional methods like alarms or manual pill boxes are not reliable, as they do not ensure whether the medicine has actually been taken. This highlights the need for a smarter and more reliable system.

Problem Statement and Motivation:

The main problem addressed in this project is the lack of an effective system to ensure timely and confirmed medication intake. Existing solutions provide reminders but do not verify whether the user has taken the medicine, leading to missed doses and health risks. The motivation behind this project is to develop a system that not only reminds users but also confirms medicine intake using sensors, thereby improving medication adherence and reducing health complications.

Overview of the Concepts Used :

The system is based on embedded system design using an Arduino Uno as the main controller. A Real-Time Clock (RTC) module is used to maintain accurate timing for medication schedules. Alerts are provided using a buzzer, LED, and LCD display. A touch sensor is used for user interaction to open the pill compartment using a servo motor. An IR sensor is used to detect whether the medicine is taken, enabling missed-dose detection. These components work together to create an automated and user-friendly smart pill box system.

CHAPTER 2

LITERATURE REVIEW

Taking medicine regularly is an essential part of maintaining good health, especially for elderly individuals and patients who are managing chronic illnesses. However, in real life, many people struggle to follow their medication schedule properly. Forgetfulness, busy daily routines, and the complexity of taking multiple medicines at different times often lead to missed or delayed doses. Over time, this can reduce the effectiveness of treatment and may even lead to serious health complications. Because of this, ensuring proper medication adherence has become an important challenge in healthcare.

Over the years, several solutions have been developed to address this problem. Traditional methods such as alarm clocks, written schedules, and manual pill boxes are simple but not very reliable, as they depend entirely on the user's memory and discipline. More advanced solutions like mobile applications provide reminders, but they require smartphones and may not be suitable for elderly users who are not comfortable with technology. There are also smart pill dispensers with features like Bluetooth, RFID tracking, or internet connectivity, but these systems are often expensive, complex to set up, and not easily accessible to everyone.

Another major limitation of most existing systems is that they only provide reminders and do not confirm whether the medicine has actually been taken. This gap between reminder and confirmation reduces the effectiveness of such systems and still leaves room for missed doses. Therefore, there is a clear need for a simple, affordable, and reliable solution that not only reminds users but also verifies medicine intake.

To overcome these limitations, the **Embedded Smart Pill Box with Missed-Dose Alert Mechanism** is proposed. This system is designed using an Arduino Uno as the main controller along with a Real-Time Clock (RTC) module to maintain accurate timing. The system is programmed to provide reminders at specific times of the day such as morning, afternoon, and evening. At each scheduled time, the device activates a buzzer, LED indicator, and displays a message on an LCD screen to notify the user.

To make the system interactive and user-friendly, a touch sensor is used. When the alert is triggered, the user must touch the sensor, which activates a servo motor to open the pill compartment. This ensures that the user actively responds to the reminder. In addition to this, an IR sensor is placed to detect whether the medicine has been taken. When the pill box is removed from the sensor's position, it confirms that the user has taken the medicine. If the medicine is not taken within a specific time, the system can identify it as a missed dose.

The main objective of this project is to provide a practical and easy-to-use solution that improves medication adherence without requiring internet connectivity or complex setup. The system is designed to be low-cost, portable, and suitable for everyday use, especially for elderly individuals. By combining reminder mechanisms with real-time detection, this project offers a more reliable approach compared to existing systems.

In conclusion, this project demonstrates how embedded systems and sensor technologies can be effectively used to solve real-world healthcare problems. By studying and analyzing existing solutions, this system has been designed to be more efficient, user-friendly, and accessible, making it a valuable tool for improving daily medication management.

CHAPTER 3

SYSTEM DESIGN

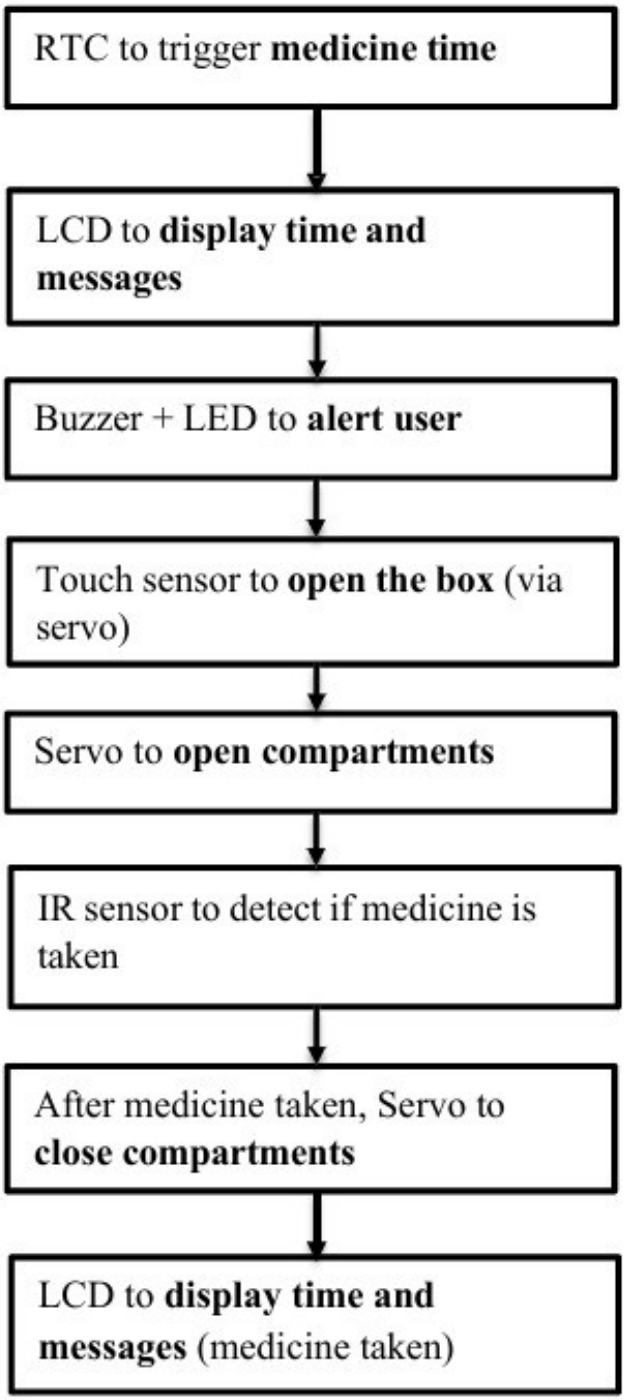
WORKING PRINCIPLE:

The system operates based on time scheduling and sensor-based interaction. A Real-Time Clock (RTC) module continuously keeps track of time and triggers alerts at predefined intervals such as morning, afternoon, and evening. When the scheduled time is reached, the Arduino Uno activates a buzzer and LED, while the LCD displays the corresponding reminder message.

Once the alert is given, the user interacts with the system using a touch sensor. When touched, the servo motor is activated, which opens the pill compartment. At the same time, an IR sensor monitors the presence of the pill box placed in front of it. When the pill box is removed, it confirms that the medicine has been taken. If the pill is not taken within a certain time, the system identifies it as a missed dose.

ARCHITECTURE:

PRODUCT WORKING FLOWCHART:



Advantages:

- Provides accurate and timely medication reminders.
- Detects whether the medicine is actually taken.
- Simple and user-friendly design suitable for elderly users.
- Low-cost and does not require internet connectivity.

Applications:

- Daily medication management for patients.
- Elderly care and home healthcare systems.
- Hospitals and rehabilitation centers..

Future Scope:

- Integration with mobile apps for remote monitoring.
- SMS or IoT-based alerts for caregivers.
- Voice-based reminders for better accessibility.
- Data logging for tracking medication history.
- Expansion to multiple compartments for different medicines.

CHAPTER 4

IMPLEMENTATION

The **Embedded Smart Pill Box with Missed-Dose Alert Mechanism** is designed to assist users in taking their medication on time through an automated and interactive system. The project is implemented using an Arduino Uno as the main controller, integrated with sensors and output devices to provide reminders and confirm medicine intake. The system is compact, cost-effective, and easy to use, making it suitable for daily healthcare applications, especially for elderly individuals.

The core working of the system is based on time scheduling and sensor interaction. A Real-Time Clock (RTC) module is used to maintain accurate time and trigger alerts at predefined intervals such as morning, afternoon, and evening. When the scheduled time is reached, the Arduino activates a buzzer and LED, and displays a reminder message on the LCD screen. This ensures that the user is clearly notified about the medication time.

To enhance user interaction, a touch sensor is used. When the alert is triggered, the user must touch the sensor, which activates a servo motor to open the pill compartment. This provides a physical action that guides the user to take the medicine. Additionally, an IR sensor is used to detect whether the medicine has been taken by monitoring the removal of the pill box placed in front of it. If the pill is removed, the system confirms that the medicine has been taken; otherwise, it can indicate a missed dose.

This implementation combines hardware components and embedded programming to create a reliable and user-friendly system. The project can be further enhanced by adding features such as mobile notifications, IoT-based monitoring, and data logging to improve functionality and usability in real-world applications.

CHAPTER 5

RESULT AND DISCUSSION

Presentation of the Final System

The final **Embedded Smart Pill Box with Missed-Dose Alert Mechanism** is a compact and user-friendly device that integrates an Arduino Uno, RTC module, LCD display, buzzer, LED, touch sensor, IR sensor, and a servo motor into a single functional system. The device is designed to provide real-time medication reminders and ensure medicine intake. At scheduled times, the system alerts the user through audio and visual signals, and the pill compartment opens using a servo motor upon user interaction. The components are neatly arranged within a compact enclosure, making the system reliable and suitable for daily use.

Evaluation of the Project's Success

The project successfully achieves its objective of improving medication adherence by providing timely reminders and confirming medicine intake. The RTC ensures accurate scheduling, while the buzzer, LED, and LCD effectively notify the user. The integration of touch and IR sensors allows both interaction and verification, making the system more reliable than traditional reminder methods. The overall implementation is cost-effective, simple, and practical, demonstrating the usefulness of embedded systems in healthcare applications.

Challenges Faced During Development

During development, several challenges were encountered, such as ensuring accurate detection using the IR sensor and properly calibrating it for reliable performance. Interfacing multiple components like the RTC, LCD, servo motor, and sensors required careful wiring and programming. Managing timing and synchronization between alerts and user interaction was also a challenge. Additionally, fitting all components into a compact and organized design required effort. These issues were resolved through testing and adjustments, improving the system's overall performance and reliability.

CONCLUSION

The **Embedded Smart Pill Box with Missed-Dose Alert Mechanism** provides an effective and practical solution to improve medication adherence. By integrating components such as an Arduino Uno, RTC module, sensors, and alert systems, the device ensures timely reminders and confirms whether the medicine has been taken. This combination of reminder and detection makes the system more reliable than traditional methods.

Throughout the development and testing, the system demonstrated its ability to assist users—especially elderly individuals and patients with memory issues—in managing their daily medication routine. Features such as audio-visual alerts, LCD display, touch-based interaction, and automatic compartment opening using a servo motor make the device simple and user-friendly. The inclusion of an IR sensor further enhances the system by verifying medicine intake, reducing the chances of missed doses.

The device is low-cost, compact, and does not require internet connectivity, making it suitable for real-world applications such as home healthcare, hospitals, and elderly care environments. It helps reduce dependency on manual reminders and provides a more consistent and stress-free medication management system.

Although the system performs effectively, there is scope for further improvement. Future enhancements may include mobile or IoT-based notifications, voice assistance for better accessibility, data logging for tracking medication history, and expansion to multiple compartments for different medicines. With further development, this system has the potential to become a more advanced and widely used healthcare solution.

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APPENDICES

Source Code:

```
#include <Wire.h>
#include <RTClib.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>

// ---------------- PIN DEFINITIONS ----------------
#define BUZZER_PIN 6
#define LED_PIN 3
#define TOUCH_SENSOR_PIN 2
#define SERVO_PIN 9
#define IR_SENSOR_PIN 5

// ---------------- OBJECTS ----------------
RTC_DS3231 rtc;
LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo lidServo;

// ---------------- VARIABLES ----------------
unsigned long startTime = 0;
bool morningMed = false;
bool afternoonMed = false;
bool nightMed = false;
bool alarmActive = false;
String currentMedicineTime = "";

// ---------------- SETUP ----------------
void setup() {
  Serial.begin(9600);

  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(TOUCH_SENSOR_PIN, INPUT);
  pinMode(IR_SENSOR_PIN, INPUT);

  lidServo.attach(SERVO_PIN);
  lidServo.write(0); // Box closed
```

```
  lcd.init();
  lcd.backlight();
  lcd.print("Med Box Active");

  if (!rtc.begin()) {
    lcd.setCursor(0, 1);
    lcd.print("RTC Error!");
    while (1);
  }

  startTime = millis();
  delay(2000);
}

// ---------------- LOOP ----------------
void loop() {

  DateTime now = rtc.now();

  // Display time when no alarm
  if (!alarmActive) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Time: ");
    lcd.print(now.hour());
    lcd.print(":");
    lcd.print(now.minute());

    lcd.setCursor(0, 1);
    lcd.print("Date: ");
    lcd.print(now.day());
    lcd.print("/");
    lcd.print(now.month());
  }

  unsigned long elapsed = (millis() - startTime) / 1000;

  // Demo timings (for testing)
  if (!morningMed && elapsed >= 60) {
```

```
    triggerAlarm("Morning");
    morningMed = true;
  }

  if (!afternoonMed && elapsed >= 120) {
    triggerAlarm("Afternoon");
    afternoonMed = true;
  }

  if (!nightMed && elapsed >= 180) {
    triggerAlarm("Night");
    nightMed = true;
  }

  // Touch to open box
  if (alarmActive && digitalRead(TOUCH_SENSOR_PIN) == HIGH) {

    openBox();

    lcd.clear();
    lcd.print("Take Medicine");

    // Check if pill taken
    if (digitalRead(IR_SENSOR_PIN) == HIGH) {
      lcd.setCursor(0, 1);
      lcd.print("Taken");
    } else {
      lcd.setCursor(0, 1);
      lcd.print("Missed Dose");
    }

    delay(2000);
    closeBox();
    stopAlarm();
  }

  delay(200);
}
```

```
// ---------------- FUNCTIONS ----------------

// Activate alarm
void triggerAlarm(String time) {
  alarmActive = true;

  lcd.clear();
  lcd.print(time + " Time!");

  digitalWrite(LED_PIN, HIGH);
  tone(BUZZER_PIN, 1000);
}

// Stop alarm
void stopAlarm() {
  alarmActive = false;

  digitalWrite(LED_PIN, LOW);
  noTone(BUZZER_PIN);
}

// Open box
void openBox() {
  for (int i = 0; i <= 90; i++) {
    lidServo.write(i);
    delay(15);
  }
}

// Close box
void closeBox() {
  for (int i = 90; i >= 0; i--) {
    lidServo.write(i);
    delay(15);
  }
}
```

EVALUATION SHEET

Reg.No : URK24CS1002

Name: BERTIL ANTO I

Course code: 23CS2050

Course Name: Embedded Systems

S.No	Rubrics	Maximum Marks	Marks
1	Industrial Certification	5	
2	Problem Statement and Objective	5	
3	Innovation and Design	10	
4	Presentation and viva	10	
5	Report	10	
Total		40	

Signature of the Faculty-in-charge